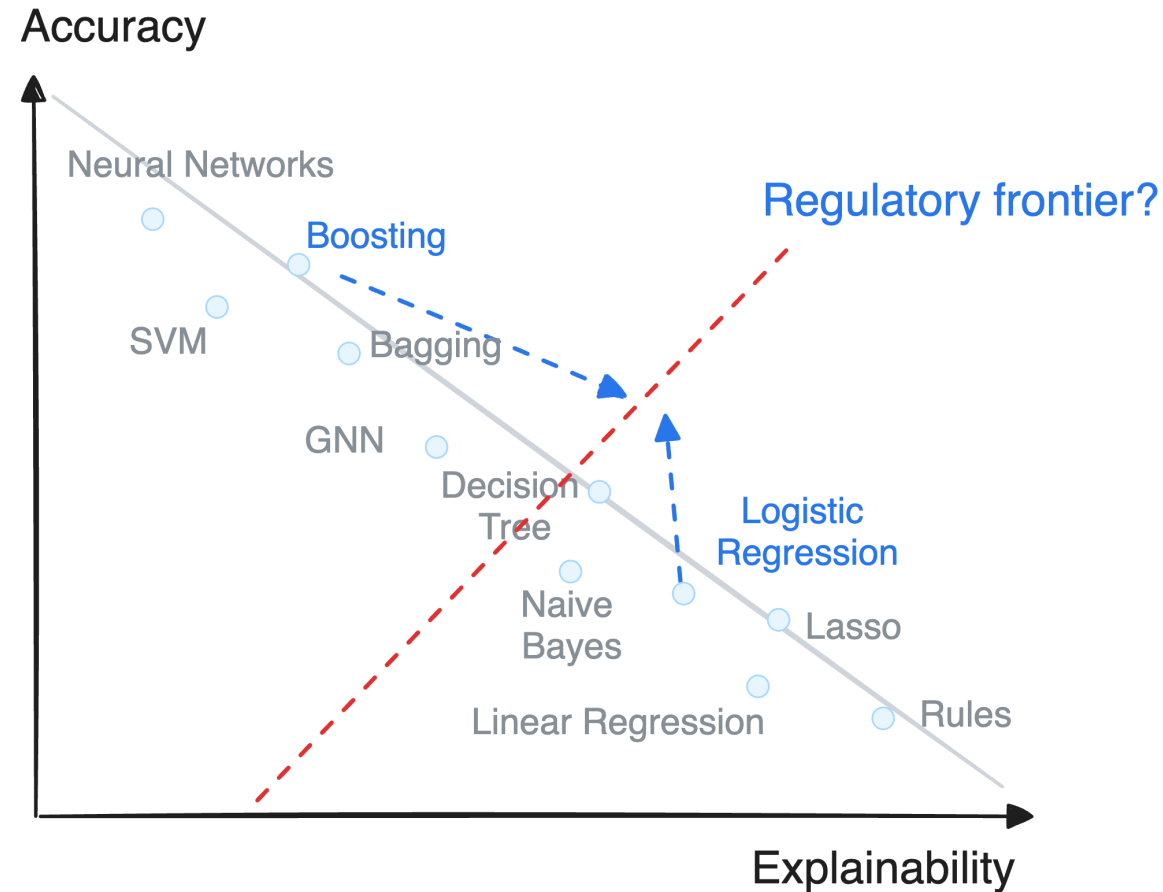


# Machine Learning in Internal Credit Risk Models



Challenges and opportunities of using ML techniques in regulated contexts

# Contents

1. Regulatory Background
2. Machine Learning in the Context of IRB Models
3. Challenges
4. Opportunities
5. Outlook for the Future

# EBA's View on Machine Learning in IRB



EBA on Machine Learning (ML) in the context of IRB (EBA/REP/2023/28<sup>1</sup>):

- Applicable for risk differentiation phase of IRB model development
- Explainability of ML models remains an open challenge
- Prudential governance around material model changes is unclear
- Competency gaps exist when it comes to new modelling techniques

Challenges related to validation of ML models have not been fully addressed in EBA's validation handbook (EBA/REP/2023/29<sup>2</sup>) as of today

1. <https://www.eba.europa.eu/publications-and-media/press-releases/eba-publishes-follow-report-use-machine-learning-internal>

2. <https://www.eba.europa.eu/activities/single-rulebook/regulatory-activities/model-validation/supervisory-handbook-validation>

# Do we really need ML for IRB?

### 'Just Say No' to Machine Learning in IRB Models

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Professor of Law at the American University  
Washington College of Law

Posted 6 December 2021

Time to read 4 Minutes

OBLB categories [Financial Regulation](#)

OBLB types [Research](#)

OBLB keywords [Financial regulation](#), [Artificial intelligence](#),  
[Financial stability](#), [European Banking Authority](#)

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More From: [Hilary J Allen](#)

In November, the European Banking Authority (EBA) issued a *Discussion Paper on Machine Learning for IRB Models*. As described therein, the aim of the Discussion Paper 'is to understand the challenges and opportunities coming from the world of machine learning (ML) should they be applied in the context of internal ratings-based (IRB) models to calculate regulatory capital for credit risk.'

As I explore in my new book *Driverless Finance*, authorizing machine learning algorithms for use in IRB models is as a very bad idea. While machine learning may have some useful internal applications for banks assessing credit risks, a

<https://blogs.law.ox.ac.uk/business-law-blog/blog/2021/12/just-say-no-machine-learning-irb-models>

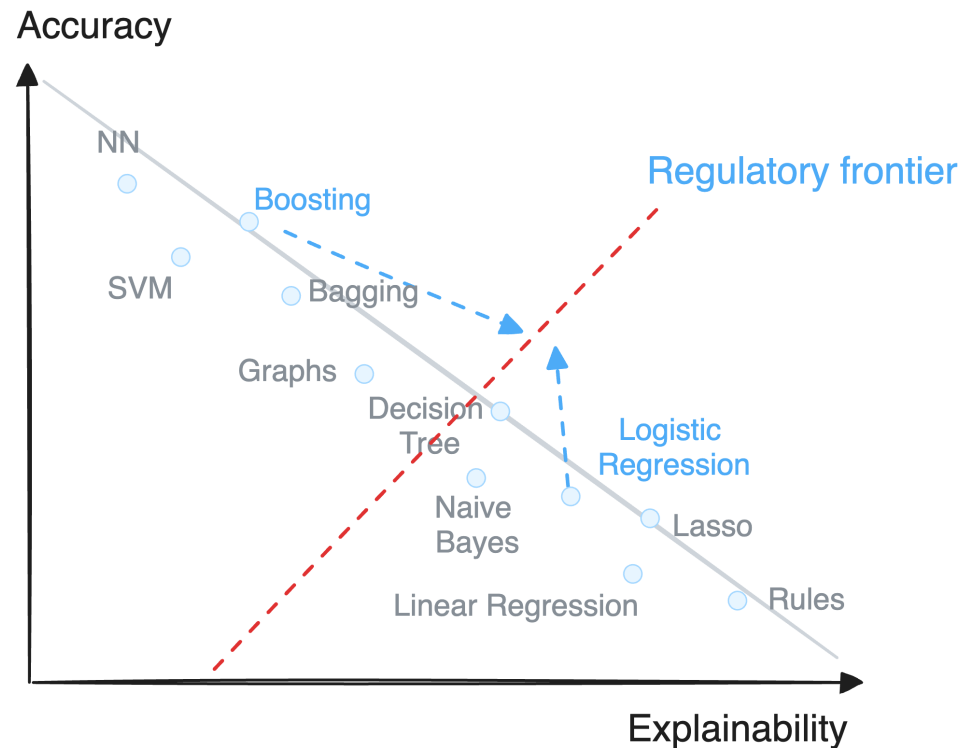
Hillary J. Allen, professor a Professor of Law, published a strong piece<sup>1</sup> as a follow up to the discussion paper by EBA (EBA/DP/2021/04)

"Explainability doesn't tell us much about how the model as a whole works or how it is likely to make decisions in an uncertain future—which is what regulators really need to know when assessing IRB models", Allen writes

The choice of using "black-box" machine learning for regulatory capital is not fully in line with the goals of financial stability, and the choice of complexity over explainability is not warranted

# Accuracy and Explainability in ML

### Accuracy - Explainability "Trade-Off"



Machine Learning (ML) models allow to gain benefits in model performance in scoring tasks by leveraging upon more flexible statistical modeling techniques

Boosting algorithms, such as gradient boosted decision trees (GBDT) have proven to be the best algorithms for tabular data outperforming logistic regression, especially on non-linearly separable data

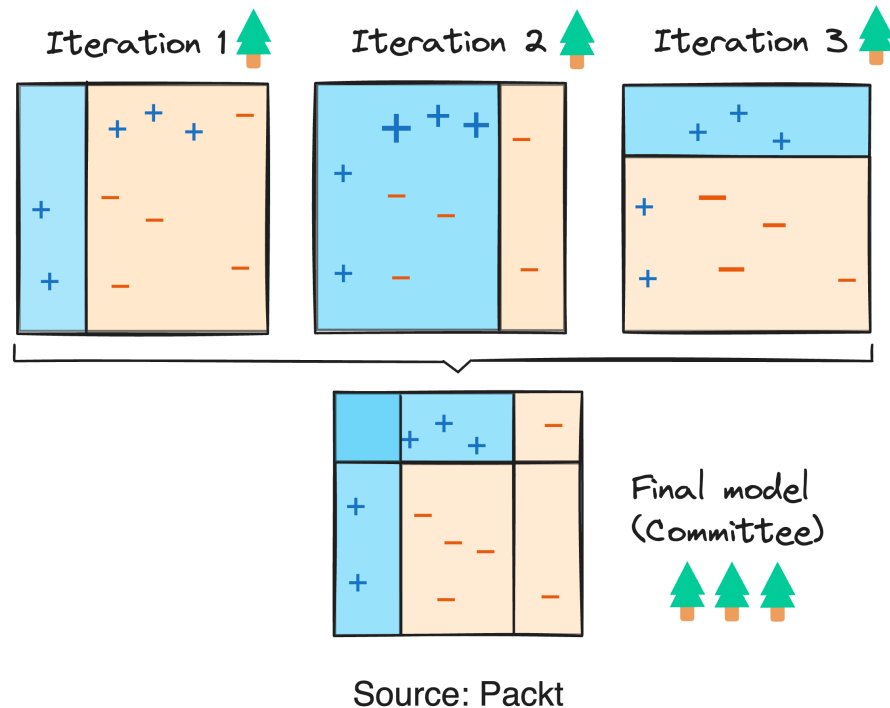
Gradient boosted tree framework for classification is based on logistic regression and performs the same task<sup>1</sup>

Source: Alexandre Duval, Paul Edwards

1. Friedman, J., T. Hastie, R. Tibshirani. Additive Logistic Regression: a Statistical View of Boosting. Annals of Statistics, 2000.

# Why Trees Outperform Simpler Models?

## Gradient boosted decision trees (GBDT)



Decision trees in gradient boosted decision tree (GBDT) models provide a flexibility that linear alternatives like logistic regression lack

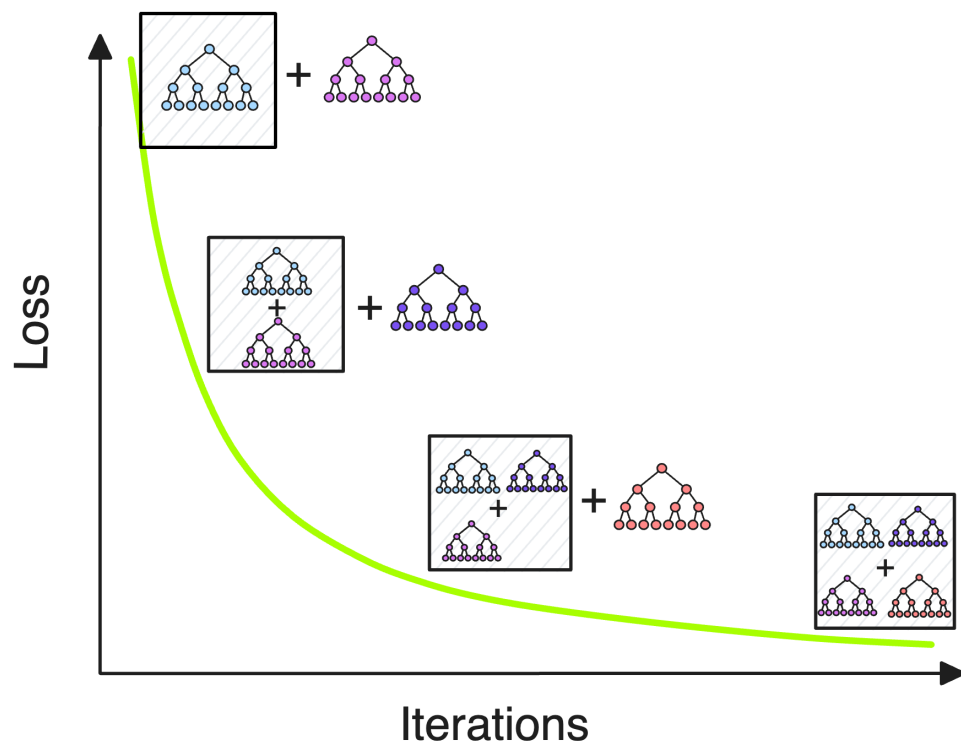
Each tree works with the errors of the previous iterations and updates its forecasts as more trees are added to the ensemble to increase accuracy

The process is somewhat similar to the Weight of Evidence (WOE) technique used by retail credit risk modellers in that we are combining evidences from different sources, in this case from different decision trees<sup>1</sup>

1. Paul Edwards et al. Real AdaBoost: Boosting for Credit Scorecards and Similarity to WOE Logistic Regression. SAS Paper 1323-2017.

# Understanding Gradient Boosting

## Gradient boosting



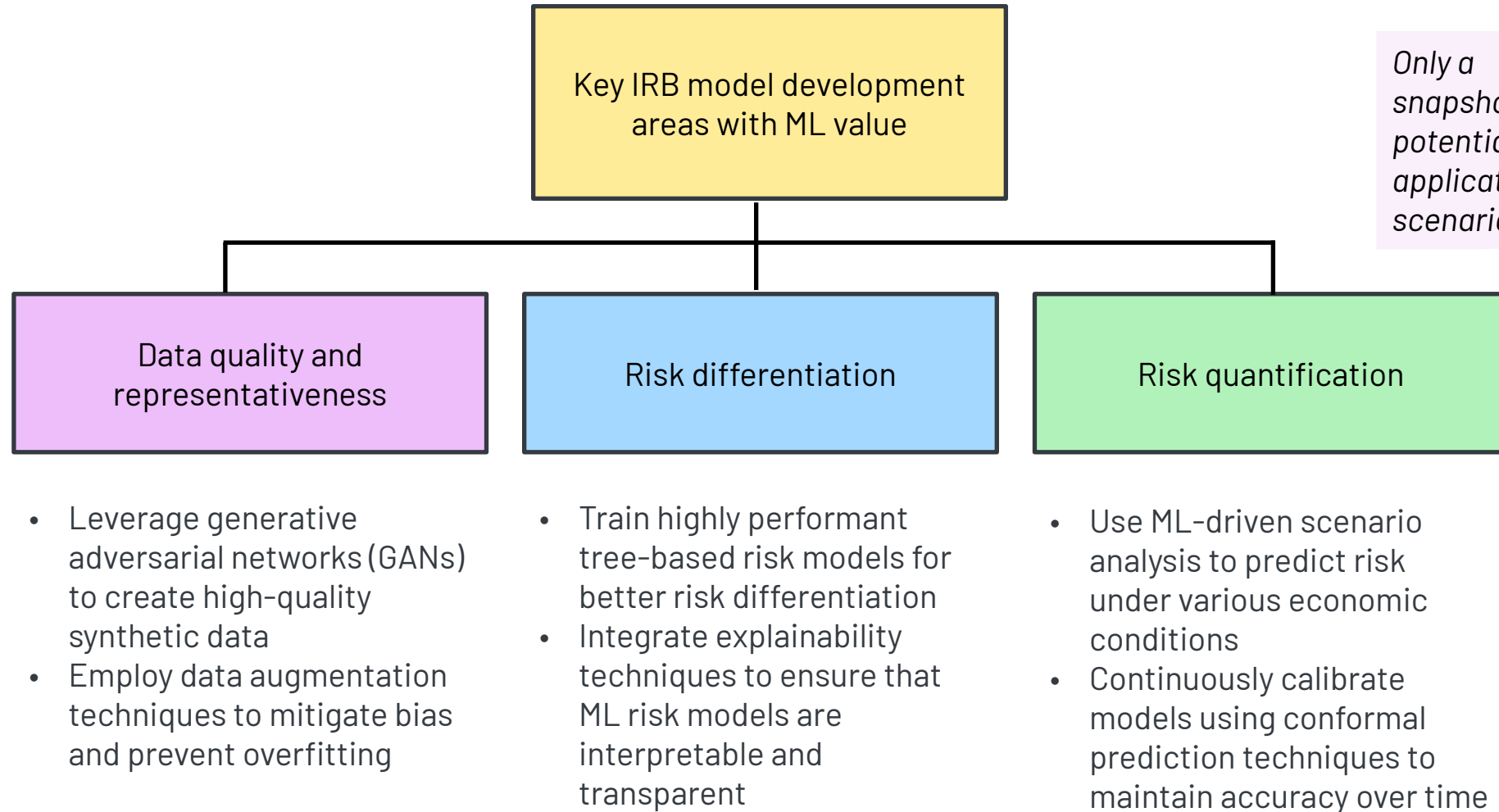
The process of gradient boosting can be understood as the estimation not on parameter level (as modellers do with linear and logistic regression), but in the functional space

Each tree's prediction is added to the intercept (base score) cumulatively to produce a probability or a linear prediction

For regulatory use, however, these models cannot be used as is. The way these models process categorical data in major libraries (XGBoost, LightGBM, CatBoost) requires a careful engineering of model constraints<sup>1</sup>

1. Monotonicity and interactions constraints to preserve the monotonicity with regards to the default rate (EBA/REP/2023/29).

# Key Improvement Areas for ML Use in IRB models

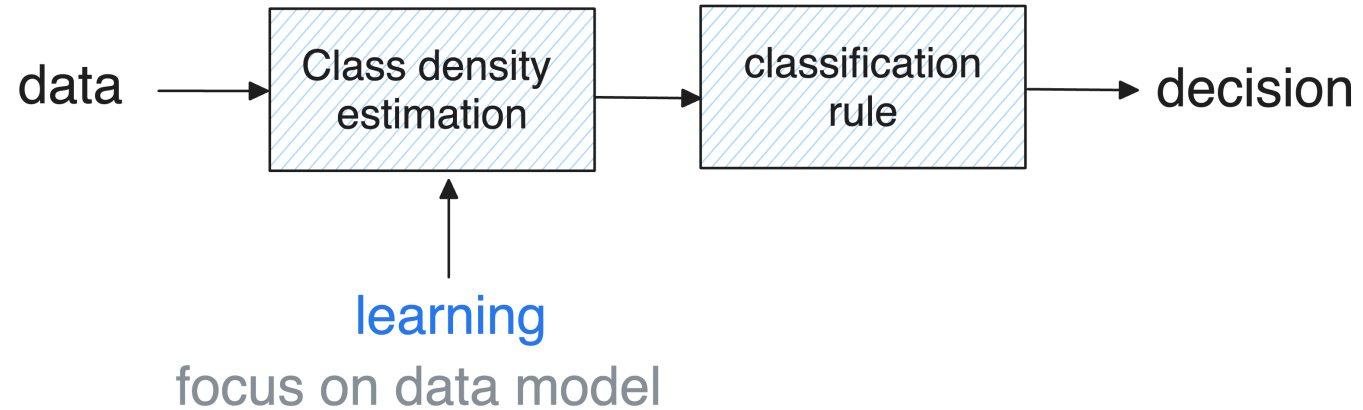
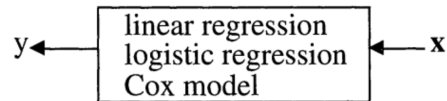


1. Use cases are selected based on a personal opinion and conversations with adopters and providers in Financial Services.

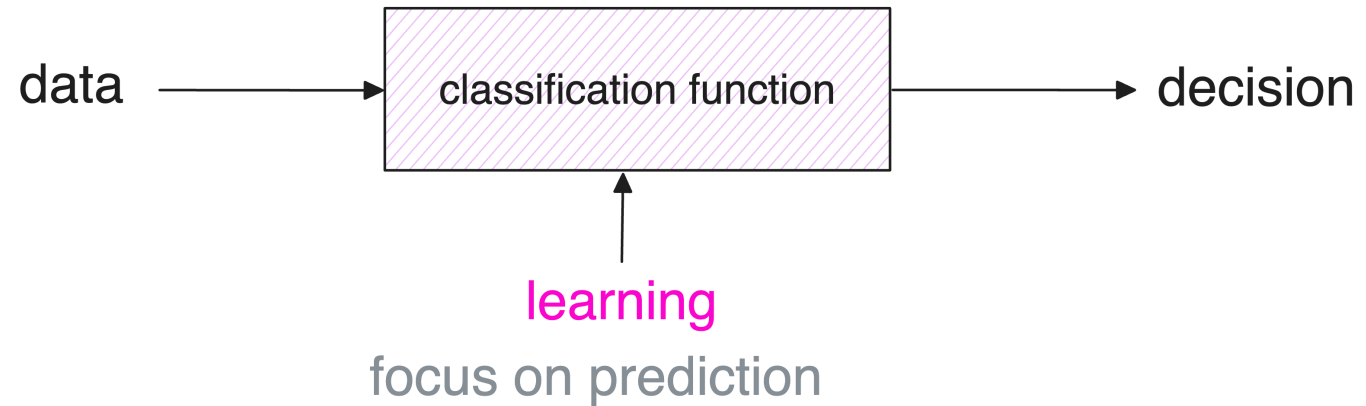
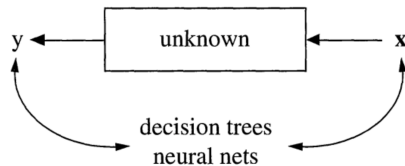


# The Explainability ("Black-Box") Problem<sup>1</sup>

Focus of statistical modeling



Focus of machine learning



1. Breiman, Leo. Statistical Modeling: The Two Cultures. Statist. Science, August 2001.

# Explainability as a Regulatory Constraint?



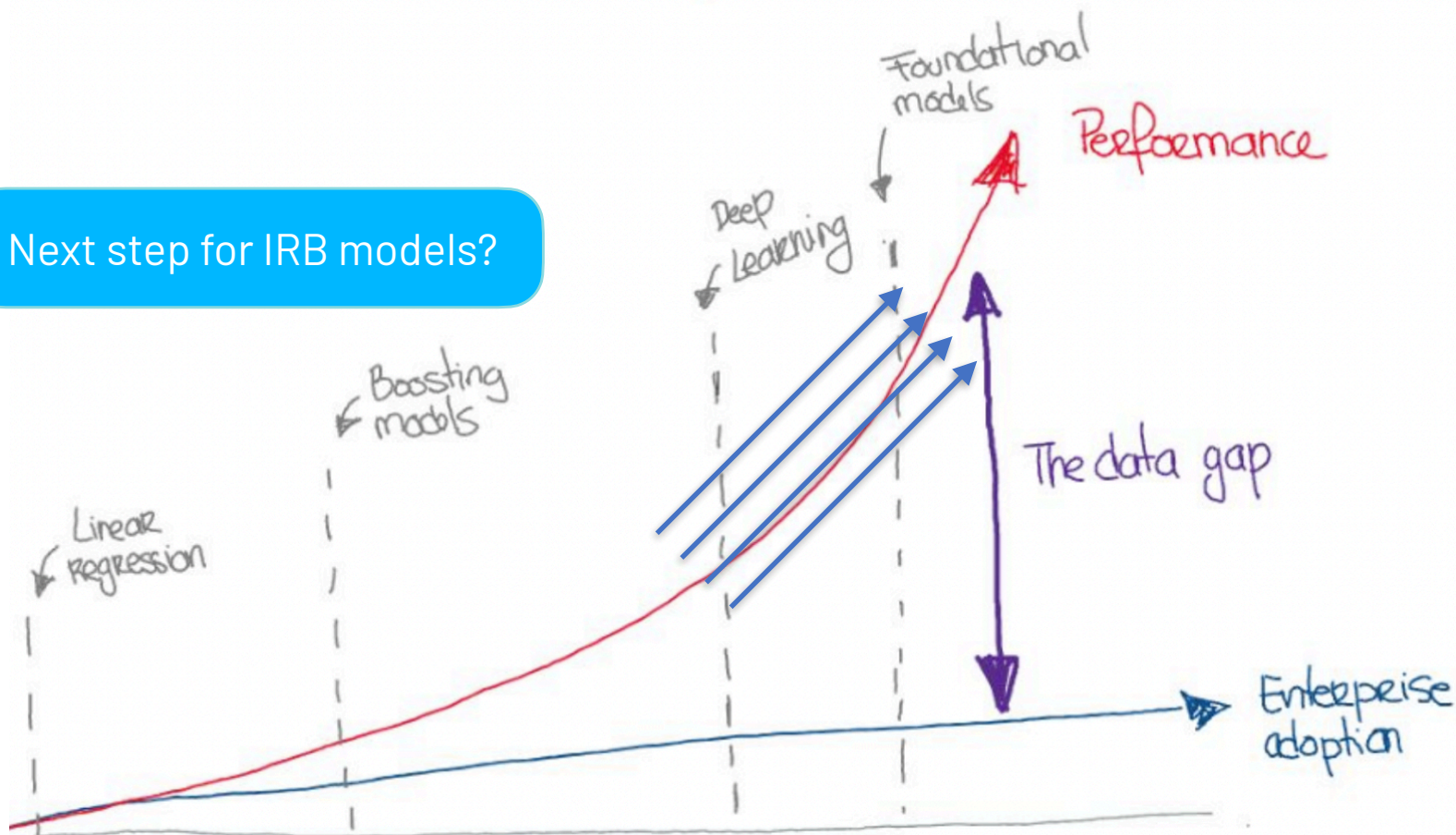
The EU AI Act in conjunction with GDPR create a regulatory framework for explainability guardrails when it comes to machine learning (ML) models

To comply, organizations must implement measures to make their models explainable, especially for high-risk cases (e.g., credit). This involves techniques such as feature importance and error analysis, post-hoc explanation methods to provide insights into how decisions are made

By adhering to these explainability requirements, organizations not only meet regulatory standards but also enhance the overall reliability and ethical standards of their AI systems

# Changing Model Landscapes<sup>1</sup>

Next step for IRB models?



## Towards the Adoption of ML in the Financial Services

- The adoption of Machine Learning (ML) in the area of Financial Services (FS) will continue, and FS may be the most impacted industry<sup>1</sup>
- As more Financial Services players adopt cloud technology, ML will play an ever-growing role in the provision of real-time analytical services to customers
- For institutions with large retail portfolios, IRB models will likely be updated with newer algorithms, especially due to their ability to include more data sources and product segments into one model and process interactions natively
- Regulatory understanding of technological trends and compliance assessments will remain key to the continuous advancements in this area in the EU

***ML solutions will provide more accurate risk assessments, helping institutions mitigate potential financial losses, but will require more human-in-the loop controls***

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1. Financial Services Skills Commission. Bridging the Skills Gap in a Rapidly Evolving Sector - Skills for the Future of Financial Services 2024.

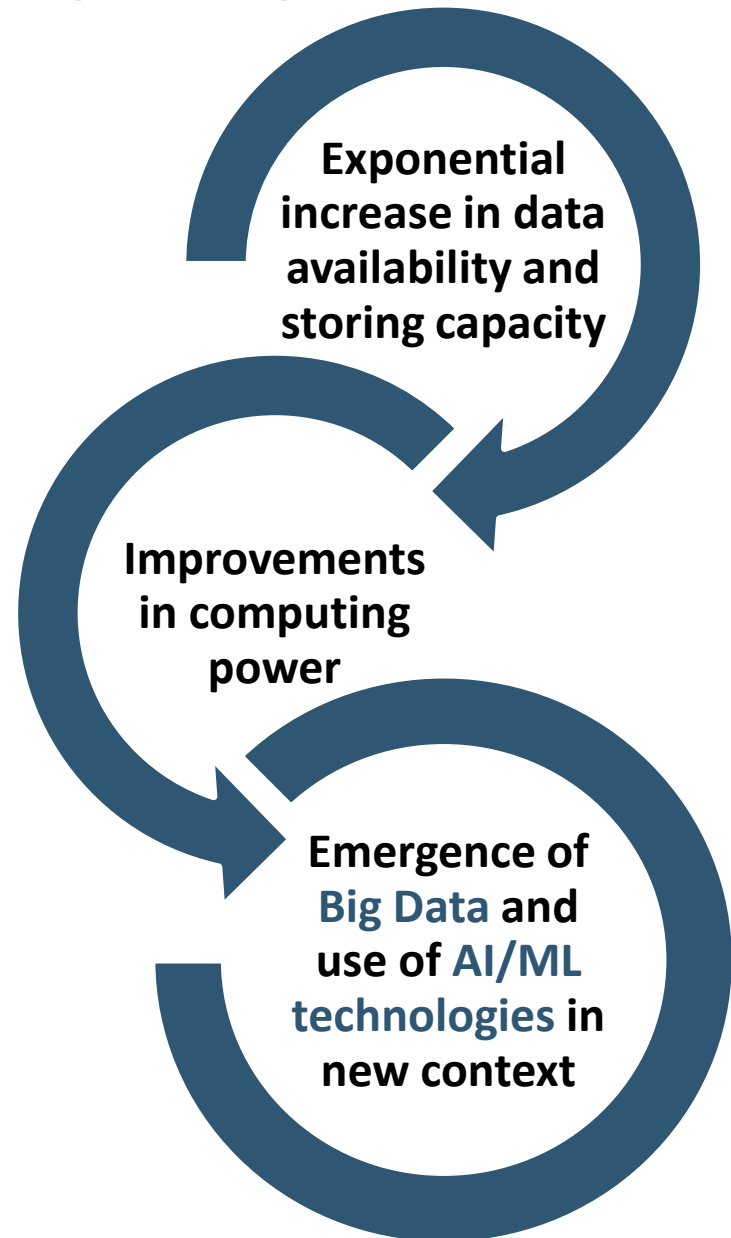
# Appendix



# Discussion paper on machine learning for IRB models

Public hearing – 16 December 2021

# Digital age supervisory challenges from AI/ML

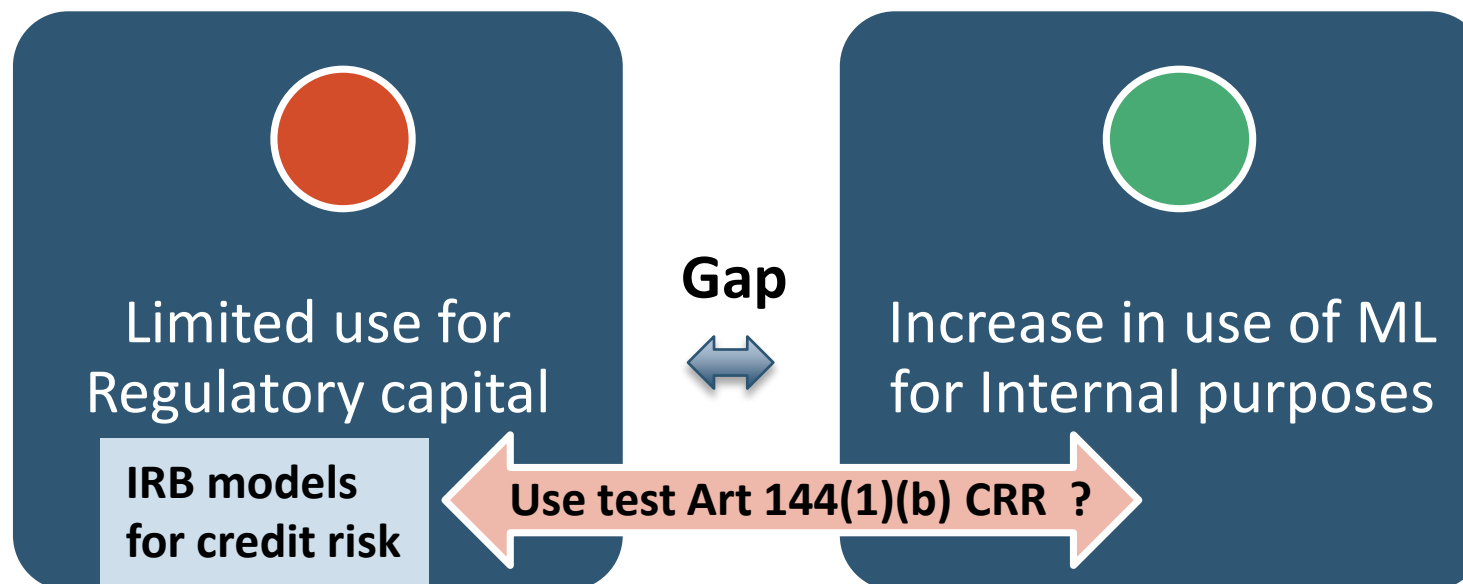


The EBA work identifies the supervisory challenges to adapt to the technological developments and provides related guidance:

- Jan 2020, EBA report on the recent trends of big data & advanced analytics (BD&AA) in the banking sector.
- April 2021, Commission legislative proposal on AI → AI used for evaluating the creditworthiness of natural persons among the high-risk use cases.
- June 2021, EBA report analysing the current RegTech landscape in EU.
- 11 Nov 2021, EBA published the **Discussion paper on ML for IRB model (DP)** for 3 months consultation.

# Background

- Why we need this discussion paper?



- What is the aim?
  - Investigate the sources of this gap and
  - discuss how ML models can coexist and adhere to the CRR requirements for IRB models for credit risk
- **Scope/definition:** more complex ML models that are difficult to interpret
- **Why stakeholder feedback is important?** Strengthening supervisory experience on ML models used in IRB



# Structure of the discussion paper

## Section 2

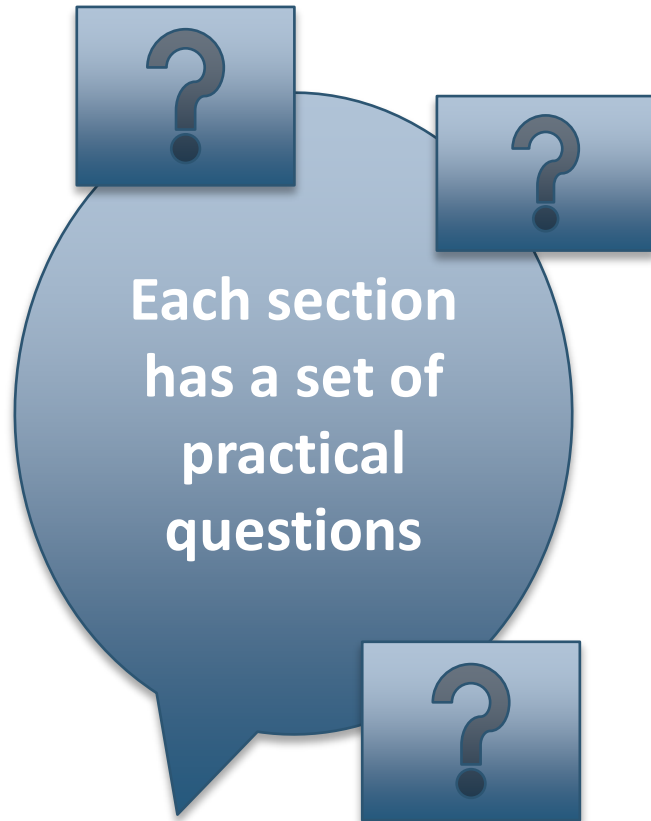
- Set the scope (complex models)
- Current use

## Section 3

- Challenges and potential benefits of using ML in developing CRR compliant IRB Models

## Section 4

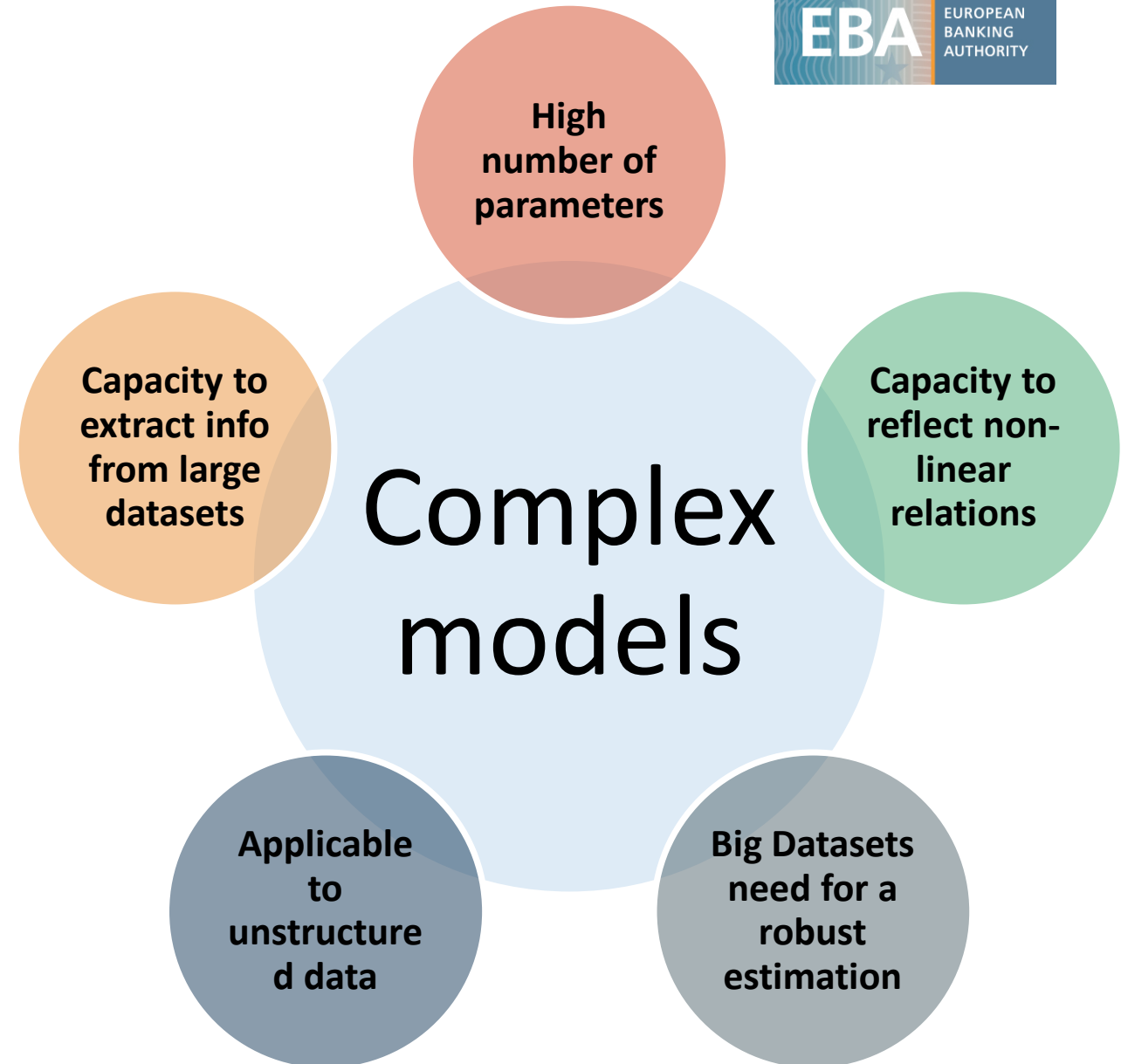
- Main concerns
- Principle-based recommendations



Each section  
has a set of  
practical  
questions

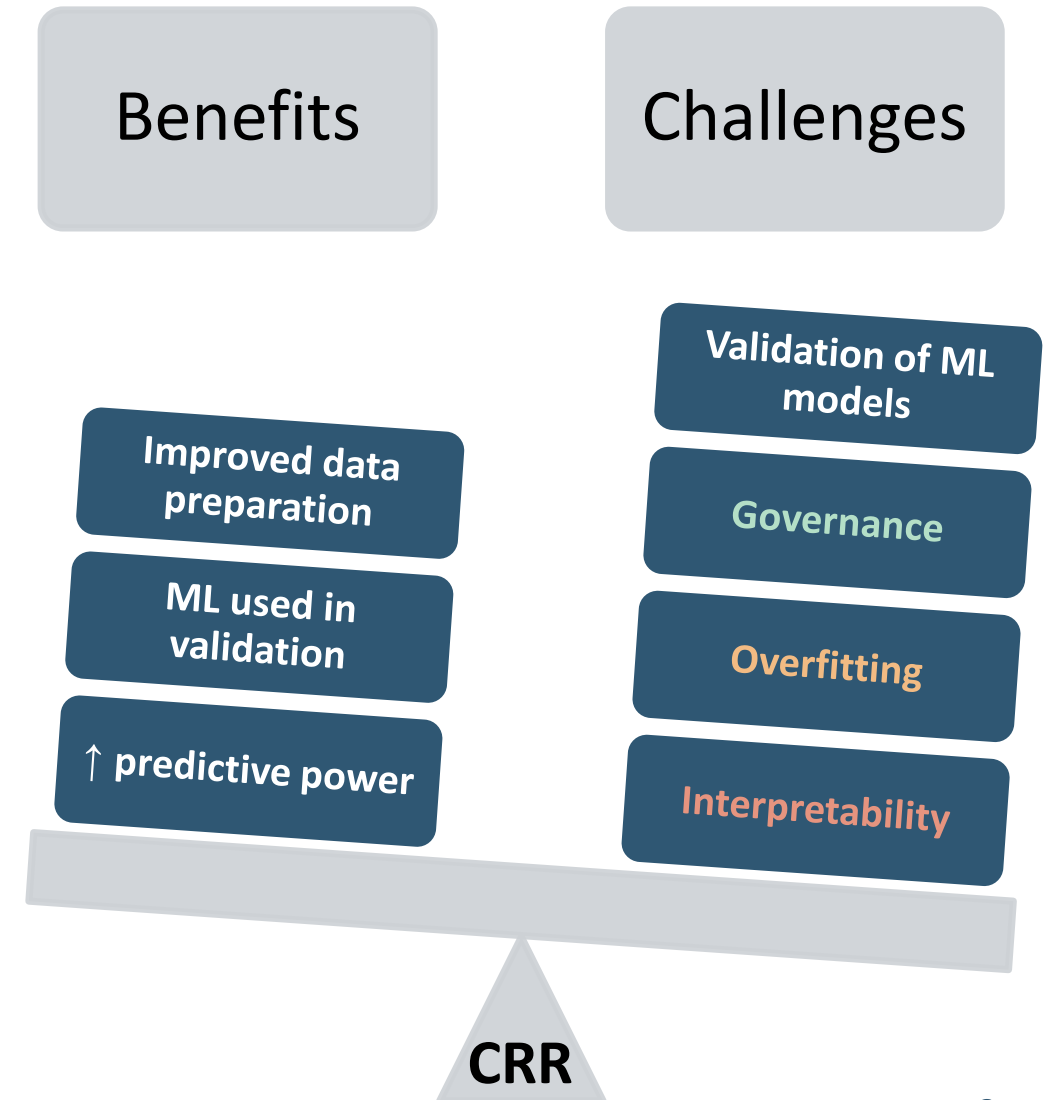
# Setting the scope

- ML covers a wide range of models with different level of complexity (from linear regression to deep neural network)
- The discussion paper refers only to the more complex that are difficult to interpret and justify to supervisors
- No clear-cut distinction between simple and complex models
- The DP provides **characteristics to consider to evaluate the complexity of a model**



# Main challenges and benefits from using ML in IRB

- The DP analyses CRR provisions to identify **challenges** and **benefits** in using ML to develop IRB compliant models
- These are **specific to the context** in which ML may be used (e.g. risk differentiation or validation) and depends on whether ML does the prediction in full or it is used as a supporting tool (e.g. data cleansing)
- Main concerns:
  1. **Interpretability of results**
  2. The governance with particular reference to **ensuring adequate understanding by their users including management functions**
  3. **The difficulty in evaluating the generalisation capacity of a the models**



# Principle-based recommendations

The DP stresses economic and supervisory principles that banks should consider when seeking approval to apply ML for regulatory purposes, these are aimed at:

- **Ensure that ML models are properly understood by their users** (CRCU, validation unit and management functions)
- **Avoid unnecessary complexity** (excessive number of parameters and reliance on unstructured data) if not justified by a significant improvement in predictive power
- **Ensure model can be interpreted and documented properly** (assess statistical and economic relations of risk drivers with model predictions)
- **Enhancing** the understanding of assumptions and behaviour of the model on specific predictions when using **human judgement** in model development and application respectively.
- **Justify and monitor frequent updates** of a model
- Focus on **validation** which may require **increased depth/frequency**

# Industry feedback is sought with a set of 17 questions

**Q 1-2**

- Explore current use of ML in IRB

**Q 3-12**

- Explore specific challenges related to internal user acceptance, human judgement, overfitting, frequency of model updates.
- Challenges in using ML for IRB models stemming from the AI act

**Q 13-14**

- Explore potential benefits of using ML in IRB and,
- In particular, the use of ML for collateral valuation

**Q 15-17**

- Explore the use of explainability techniques and, the strategy to ensure understanding by the ML models users
- Investigate industry concerns around the recommendations given

# Conclusions and next steps

- ML models can provide added value in the IRB context provided that they ensure acceptable monitoring, validation and explainability of methodology and model outcomes.
- The need for competence in ML is becoming increasingly important.
- Decision to go forward with ML in IRB should be based on prudential terms but also considering other relevant aspects (e.g. Ethical or consumer protection considerations)
- The discussion paper seeks stakeholders' feedback on many practical aspects related to the use of machine learning in the context of IRB (deadline 11 February 2022)
- Need to provide further guidance will be assessed based on the feedback received
- Potential developments related to relations with the Commission AI act





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